

Liquid State

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Liquid State: Liquid State is the intermediate State between Solid and gas. It posses Fluidity like gas and incompressibility like Solid.

Properties of Liquid:

1. Shape: They have no Shape of their own but assume the Shape of the Container in which they are kept.
2. Volume: They have definite Volume Since molecules are closely packed and cohesive Force is strong.
3. Compressibility: They are incompressible Since cohesive Force is strong.
4. Diffusion: When two miscible liquids are put together, there is slow mixing as the molecules of one liquid move into the other liquid. This is diffusion in liquid. Liquid diffuse slowly as Comparison to gases Since Space available for movement of molecule is less.

Evaporation: The process of changing liquid into vapour is called evaporation. It is spontaneous phenomenon and occurs from Surface of liquid.

- Evaporation may be explained on the basis of motion of. At any given temperature all the molecules do not posses the same kinetic energy, their average energy is only same. Some molecule move slowly, Some at intermediate rate and some move very fast. A rapidly moving molecules wear

The structure of liquid may possess sufficient kinetic energy to overcome attractive force of attraction by neighbours and escape.

Features of Evaporation:

- i) Spontaneous Process
- ii) Endothermic Process
- iii) Surface phenomenon
- iv) Cause Cooling.

Evaporation Cause Cooling

Due to evaporation, the molecules which are on the surface having high kinetic energy are lost leaving behind the molecules having lower kinetic energy. This results in decreasing the average kinetic energy of liquid and may cool the liquid.

Factors affecting evaporation

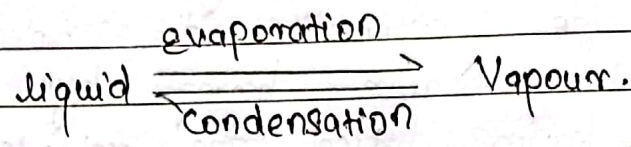
- Evaporation depends on the strength of intermolecular force (cohesive force). The liquid having low intermolecular force evaporates faster in comparison to liquid having high intermolecular force. For e.g.: Ether evaporates more quickly than alcohol and alcohol more quickly than water.

Intermolecular force of attraction, $H_2O > ROH > RO_2$
Extent of evaporation, $RO_2 > ROH > H_2O$.

- Surface area:- Evaporation is a surface phenomenon. Larger the surface area, greater is the opportunity of the molecules to escape. Thus, rate of evaporation increases with increase in surface area.

Vapour Pressure: When the space above liquid is closed, the molecules cannot escape into open but strike the walls of container, rebound and may strike the surface of the liquid where they are trapped. The return of molecule from vapour state to the liquid state is known as Condensation. As evaporation increases, the no. of molecules in Vapour state increases which increases Condensation.

When rate of Condensation becomes equal to rate of evaporation, the vapour in closed container is in equilibrium with liquid.



At equilibrium, the concentration of molecules in the Vapour phase remain unchanged. Thus Vapour pressure is defined as "The pressure exerted by vapour in equilibrium with liquid at a given temperature is called Vapour pressure."

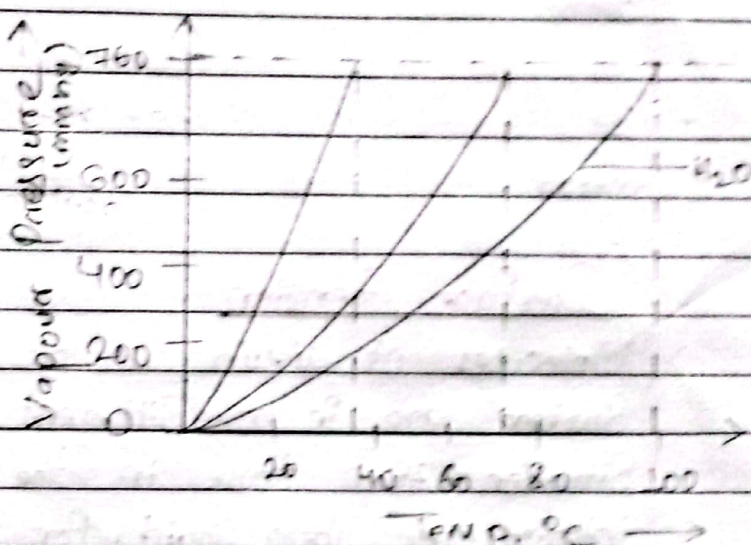
Effect of temperature on Vapour pressure:

- As the temp. of liquid is increased, the Vapour pressure also increases because at high temp. the liquid contains more molecules having high kinetic energy to overcome the intermolecular force of attraction.
- Non-polar or less polar Solvent have high Vapour pressure since they have weak intermolecular attractive force.
- Polar Solvent like H_2O have less vapour pressure.

Boiling: If the temp. of liquid is increased to

Such extent that Vapour Pressure become equal to atmospheric pressure then liquid start boiling and temp. is referred to as boiling point of liquid.

Thus, Boiling is defined as "the process of conversion of liquid into Vapour, when Vapour pressure is equal to atmospheric pressure is called boiling".



Effect of pressure on boiling:

- The b.p of liquid changes with the change in external pressure. A liquid may boil at temp. higher than normal when external pressure is greater than one and it may boil at temp. lower than normal when external pressure is less than atmospheric pressure.
- At high altitude, the atmospheric pressure is less than 760 mmHg. So water boils at lower temp. i.e. below 100°C . Thus, food cooks more slowly at high altitude.
- Food cooks faster in pressure cooker because temp. of boiling water is higher in pressure cooker i.e. more than 100°C .

Difference between Evaporation and boiling.

P.T.O.

Evaporation

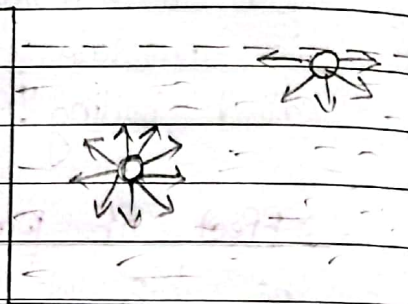
- It takes place spontaneously at all temp.
- It is surface phenomenon.

Boiling

- It takes place only at particular temp. where vapour pressure is equal to atmospheric pressure.
- It involves formation of bubbles below the surface of liquid.

Surface tension

Molecules of liquid at the surface are in a different situation than those in the interior of the liquid. For eg:- a molecule lying inside the liquid is surrounded by the other molecule and so is attracted equally in all direction. Hence, the net force of attraction acting on the molecule is zero but a molecule lying at the surface is attracted downwards by the molecule lying in the bulk of liquid. Thus, a molecule lying at the surface experiences a net inward attraction as a result of this inward pull on all molecules at the surface, the surface behaves as if it were under tension and this property of liquid is called surface tension.



Then, Surface tension of liquid is defined as the force acting at right angles to the surface along one centimetre length of surface.
Unit of Surface tension = dyne/cm (CGS)
= Nm^{-1} (SI)

Q. Define surface tension? What is the cause of surface

tension? Write CGS and SI unit of surface tension.

Different Phenomenon due to Surface tension:

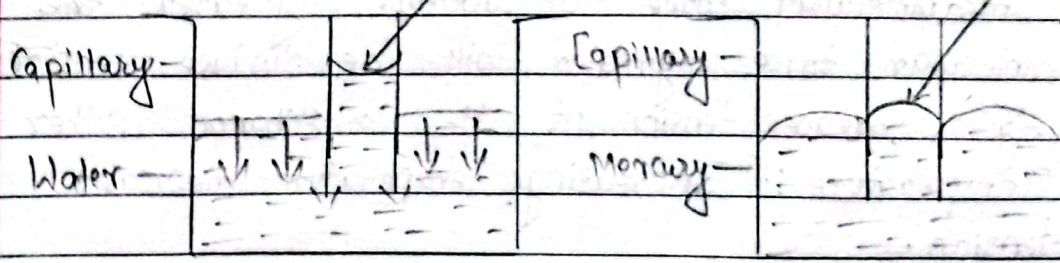
A. Liquid drops are spherical in shape, why?

→ Due to surface tension, liquid try to attain minimum surface area and for given volume, surface area is minimum for sphere. So rain drops or liquid drops are spherical in shape.

B. The rise or fall of liquid in capillary tube, why?

→ If the capillary tube is dipped into liquid, the liquid rises into the capillary tube to certain height. This phenomenon takes place because adhesive force (force of attraction between polar molecule and glass) is greater than cohesive force i.e. surface tension. The rising stops when adhesive force is balanced by weight of liquid. Thus, water level rises in glass capillary tube with concave meniscus upward.

If the capillary tube is dipped in mercury, a decrease of the mercury level occurs with convex meniscus. This is because cohesive force is greater than adhesive force between glass and mercury.



C. Efficiency of toothpaste.

→ Due to minimum surface tension, the liquid spread

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over surface to increase its surface area. The toothpaste and mouth washes consists of the substance which lowers the surface tension. Due to this, the paste spread over surface they come in contact and clean the mouth.

D. Cleansing action of Soap and detergent.

→ Soap and detergents consists of that material which lowers the interfacial tension between water and grease or dirt. This facilitates the mixing of water and dirt.

Effect of temperature on Surface tension:

Surface tension is force. So with increase in temperature, the surface tension of liquid decreases. The intermolecular force weakens when temp. is increased.

Q. Clothes are washed more efficiently in hot water than cold water?

→ Hot water has less surface tension so it spread more over clothes and clean the dirt efficiently.

Factors affecting Surface tension:

i) Intermolecular force of liquid:- greater the intermolecular force greater will be surface tension.

Eg:- Surface tension of $H_2O > Ethanol > Ether$.

ii) Temperature:- Increasing temperature decreases surface tension.

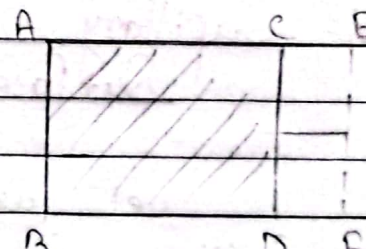
iii) Presence of impurities:- Organic molecule such as alcohol, soap, detergent concentrate at surface and decreases the surface tension of solution. NaCl increases the surface tension.

Surface Energy

To increase the area of surface, some work has to be done against the inward pull.

Consider a soap solution film contained in rectangular wire frame ABCD in which the side CD is movable. In order to extend surface area of film, movable wire has to be pulled from position CD to EF. Thus, some work has to be done against surface tension.

Thus, Surface Energy is defined as "The work done in ergs required to extend the surface area by 1 cm^2 is called Surface energy."



$$\text{Unit: - Work / cm}^2 = \frac{F \times d}{\text{cm}^2} = \frac{\text{dyne} \times \text{cm}}{\text{cm}^2} = \text{dyne cm}^{-1}.$$

Q. Water droplet on the wax paper takes spherical shape?
→ On wax or oily surface, the cohesive force (surface tension) dominates the adhesive force between polar water molecule and non-polar water molecule. So due to high surface tension, water takes spherical shape.

Interfacial tension: When two immiscible or partly miscible liquids are taken in a vessel then surface tension acting along their surface of separation is called interfacial tension. Its value is intermediate between surface tension of both liquid or sometimes less than both.

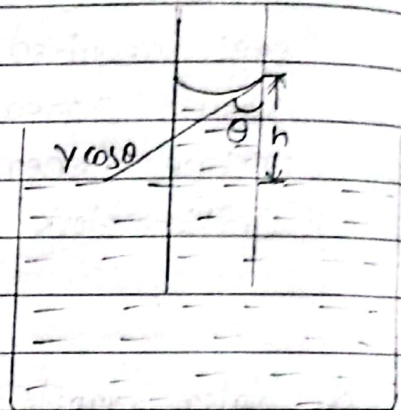
Surface active agents: There are certain substances like soap, sulphonic acid, organic compound like

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Methyl alcohol, acetone, ethyl alcohol when added to water even in small amount decreases the surface tension of water are called surface active agents. These substances decrease surface tension of water which easily wet the grease and mix it and remove it from cloth.

Capillary rise Method for determining Surface tension of Liquid

Let, us consider a fine capillary tube is put into liquid which wets the glass.
Let,



r = radius of capillary tube

γ = Surface tension of liquid.

θ = angle of contact.

h = height of liquid in capillary tube.

ρ = density of liquid.

g = acceleration due to gravity.

The force of surface tension pushes the liquid into the capillary to certain height is given by,

$$F_1 = \text{Circumference} \times \gamma \cos \theta$$

$$F_1 = 2\pi r \gamma \cos \theta \quad \text{--- (i)}$$

Weight of liquid acting downward,

$$W = m \times g$$

$$W = V \times \rho \times g$$

$$W = \pi r^2 h \rho g \quad \text{--- (ii)}$$

[i.e. Volume of cylinder = $\pi r^2 h$]

The upward force is balanced by weight of liquid at equilibrium. So,

$$2\pi r \gamma \cos \theta = \pi r^2 h \rho g$$

$$\gamma = \frac{\pi r^2 h \rho g}{2\pi r \cos \theta}$$

$$\gamma = \frac{r h \rho g}{2 \cos \theta}$$

For most of liquid, $\theta = 0$

$$\text{So, } \cos \theta = 1$$

Hence,

$$\therefore \boxed{\gamma = \frac{r h \rho g}{2}}$$

Determination of Surface tension by drop Weight Method using Stalcyrometer.

⇒ Principle :- "If a liquid be allowed to drop through a tube having Capillary at its end, the drop forming at a Capillary end is supported by the Surface tension acting upward along the Circumference of Capillary tube. So as long as Surface tension remain longer than weight of drop doesnot fall. However, when the drop grows to such a shape that its weight exceed the Surface tension the drop falls".

When the drop just falls, the force acting downward due to weight is equal to force Supporting the drop.

$$m \times g = 2\pi r \gamma \quad \text{where, } w = \text{wt. of drop.}$$

$$w = 2\pi r \gamma \quad \text{--- (1)}$$

$\gamma =$ Surface tension of liquid.

Procedure :- The Stalcyrometer is cleaned & dried and water is sucked up through rubber tube above the mark M. and meniscus is all at M with the help of pinch cork provided

With rubber tube. Then water between mark M and N is allowed to come out in form of drop. The drop are adjusted with pinch cock and no. of drop formed is counted. and above process is repeated.

Now, from eqn (1),

$W \propto \gamma$, i.e. weight of liquid drop is proportional to surface tension of liquid.

If γ_1 be the surface tension of water and γ_2 be the surface tension of liquid under consideration. Then,

$$\frac{\gamma_1}{\gamma_2} = \frac{W_1}{W_2} \quad \text{where, } W_1 = \text{wt. of one drop of water.}$$

$$W_2 = \text{wt. of one drop of liquid.}$$

If n_1 and n_2 be the no. of drop counted in same volume of water and liquid then,

$$W_1 = n_1 \times m_1 \times g \quad \text{and} \quad W_2 = n_2 \times m_2 \times g$$

$$= \frac{V}{n_1} \times d_1 \times g \quad = \frac{V}{n_2} \times d_2 \times g$$

putting value of W_1 and W_2 in above relation,

$$\frac{\gamma_1}{\gamma_2} = \frac{V \times d_1 \times g \times n_2}{n_1 \times V \times d_2 \times g}$$

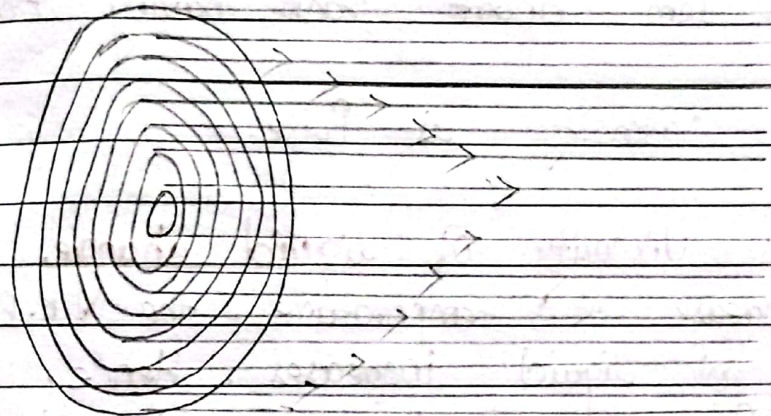
$$\boxed{\frac{\gamma_1}{\gamma_2} = \frac{n_2 d_1}{n_1 d_2}}$$

Using this relation, surface tension of liquid with respect to water can be calculated.

Viscosity

Consider, a liquid flowing through a narrow tube. All parts of liquid do not move with same velocity through the tube. Imagine the

liquid to be made up of large no. of thin cylindrical Co-axial layers. The layer which is in contact with the walls of tube is almost stationary.



As we move from wall towards the centre of tube, the velocity of cylindrical layer goes on increasing. It is maximum at the centre. Conversely, we may say that as we move from centre towards the wall, the velocity of layer keep on decreasing.

In other words, every layer offer some resistance or friction to the layer immediately below it. This "force of friction or resistance which one part of liquid offer another layer of liquid is called viscosity".

It has been found that the force of friction (F) between two cylindrical layer each having area A sq. cm separated by n cm having velocity difference of v cm/sec is given by

$$F \propto \frac{A \times v}{n}$$

i.e. $F = \eta \frac{A \times v}{n}$ Where, η is constant known as Coefficient of Viscosity.

If $n = 1$ cm, $A = 1$ sq. cm and $v = 1$ cm/sec,

then, $F = \eta$ thus,

Coefficient of viscosity is defined as "Force of friction in dyne required to maintain a velocity difference of 1 cm/sec between two parallel layers of 1 cm apart each having Area of 1 sq. cm."

- Unit of Viscosity is Poise.

How does Viscosity of liquid change with temp?
 $\Rightarrow$ With increase of temperature, the K.E. of molecules of liquid increases. Hence, liquid starts flowing faster and thus viscosity decreases.

Fluidity (ϕ) :- The reciprocal of Viscosity is called Fluidity.

i.e. $\phi = \frac{1}{\eta}$

Unit of Fluidity = Poise⁻¹.

Fluidity of liquid is measure of tendency of liquid to flow fast.

Determination of Viscosity by Ostwald's Viscometer.

Working Principle :- Viscosity of liquid is determined by Ostwald's viscometer method. This principle is based on Poiseuille's law. A/c to Poiseuille's law,
 "Different liquid flow at different rates and Volume 'V' of liquid flowing through a tube having radius 'r' and length 'l' under difference of pressure 'P' at the two end within time 't' is given by,

$$V = \frac{\pi P r^4 t}{8 \eta l}$$

$$\therefore \boxed{\eta = \frac{\pi P r^4 t}{8 V l}} \quad \text{Where, } \eta = \text{Coefficient of Viscosity.}$$

With the help of above relation, absolute value of coefficient of viscosity can be determined.

II Experimental Procedure for determination of relative Velocity of a liquid.

→ The viscometer is cleaned & dried. A definite volume of water is put into the wide limb and sucked into the bulb A till it rises above the mark 'a'. The water is now allowed to flow back through the capillary tube. Time taken by liquid to flow from mark a and b is noted with the help of stop watch. Let, time taken be equal to t_1 . The viscometer is dried and same volume of liquid is taken and time required for liquid to flow between same mark is recorded. Let it be t_2 . Suppose η_1 and η_2 represent viscosity of water and liquid then,

$$\frac{\eta_1}{\eta_2} = \frac{\pi r^4 t_1 \rho_1}{8 v l} \times \frac{8 v l}{\pi r^4 t_2 \rho_2}$$

$$\frac{\eta_1}{\eta_2} = \frac{t_1 \rho_1}{t_2 \rho_2} = \frac{t_1 h_1 d_1 g}{t_2 h_2 d_2 g}$$

$$\frac{\eta_1}{\eta_2} = \frac{d_1 \cdot t_1}{d_2 \cdot t_2}$$

Thus, viscosity of one liquid is known by knowing viscosity of water. Density of liquid is determined by specific gravity bottle.

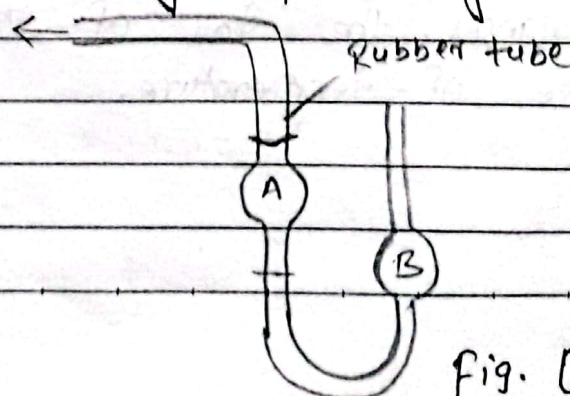


Fig. Ostwald Viscometer.

Q. Between diethyl ether and ethanol which has high viscosity?

→ Viscosity arises due to intermolecular force of attraction between liquid molecules. More be the intermolecular force higher will be viscosity. The viscosity of ethanol is higher than diethyl ether because ethanol can form intermolecular H-bonding and form aggregate mass. Whereas diethyl ether cannot form H-bonding.

Application of Viscosity :-

1. Blood pressure is caused because of strain in pumping blood through constricted arterial capillaries due to more viscous blood. A patient suffering from high blood pressure is given medicine which lower the blood viscosity.
2. In asphyxia, failure of breathing leads to accumulation of CO_2 in blood which swells blood corpuscles. A rise in viscosity of blood result heart failure. A medicine is given to patient which lower viscosity.
3. Oil used for lubricating precision instrument such as watch are mixed with long chain wicking polymer in order to overcome effect of temp. on viscosity. When temp. rises, polymer uncoil and increase viscosity. It balances the fall of viscosity of oil due to rise of temperature.